

CSA1304-THEORY OF COMPUTATION LAB EXPERIMENTS

8. Write a C program to simulate a Non-Deterministic Finite Automata (NFA) for the given language representing strings that start with b and end with a

AIM

Write a C program to simulate a **Non-Deterministic Finite Automata (NFA)** for the language representing strings that **start with 'b' and end with 'a'**.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int len;

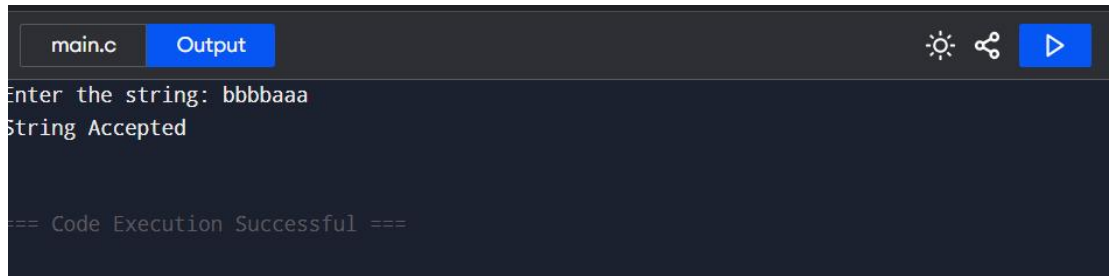
    printf("Enter the string: ");
    scanf("%s", str);

    len = strlen(str);

    if(len > 0 && str[0] == 'b' && str[len - 1] == 'a')
        printf("String Accepted\n");
    else
        printf("String Rejected\n");

    return 0;
}
```

OUTPUT



The screenshot shows a dark-themed code editor interface. At the top, there is a tab labeled 'main.c' and a blue button labeled 'Output'. To the right of the 'Output' button are three icons: a sun (theme toggle), a share icon, and a blue button with a white play icon. The main area of the editor displays the following text:

```
Enter the string: bbbbaaa  
String Accepted  
  
=== Code Execution Successful ===
```